Use the following graph for the Traveling Salesperson problem. Find the least cost path in the graph for the TSP problem.

```
   A   B   C   D   E   F   G   H
A  [0,  10, 15, 20, 25, 30, 35, 40],   # City 0
B  [12,  0,  35, 15, 20, 25, 30, 45],  # City 1
C  [25, 30,  0,  10, 40, 20, 15, 35],  # City 2
D  [18, 25, 12,  0,  15, 30, 20, 10],  # City 3
E  [22, 18, 28, 20,  0,  15, 25, 30],  # City 4
F  [35, 22, 18, 28, 12,  0,  40, 20],  # City 5
G  [30, 35, 22, 18, 28, 32,  0,  15],  # City 6
H  [40, 28, 35, 22, 18, 25, 12,  0 ]   # City 7
```

1. Use the local beam search to solve the problem. Give a comparative analysis of the algorithm when the beam width k=3,5,10. Does the convergence depend on the value of k?
2. Use the genetic algorithm to solve the problem. Use one crossover point and two crossover points to generate the offspring. Does the number of crossover points impact the convergence rate?